COMP 381 AB1
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Housing Price Predictor for Best Real Estate Deals

Topic and Motivation

Purchasing a home can be tricky, especially in the modern housing market. Prices have been volatile, and affording a home remains a challenge for Canadians (Hudes, S). For buyers, it can be difficult to tell just from a listing whether they're getting a fair price or overpaying. With that in mind, we set out to build a tool that could help take some of the guesswork out of the process. Using machine learning, our project predicts how much a home should cost based on its features, then flags it as either a "good deal" or a "bad deal".

Our Approach

We approached the problem in three main steps: classifying data, clustering data into groups, and finally, performing linear regression on the groups. Before training, we performed several preprocessing steps to maximize model performance. These included encoding binary columns numerically and one-hot encoding categorical columns. Our goal with classification was to fill in missing values to later perform training on. We trained three classification models on complete data: LDA, KNN, and Logistic Regression. These models were then compared in accuracy to determine the most suitable model to be used in the project. After filling in missing data, we then used clustering to divide data into similar areas. To avoid the curse of dimensionality, we limited clustering to geographic features, where local market conditions, neighborhood characteristics, and proximity-based factors naturally influence pricing. For each geographic cluster, we then trained a separate linear regression model, and computed the difference between the predicted and actual listing price for each property. A large positive gap was flagged as a strong deal. Based on the results, we ranked the top five deals per cluster.

References

Hudes, S. (2026). *TD revises 2026 housing market forecast.* CBC News. https://www.cbc.ca/news/business/td-revised-housing-market-forecast-9.7143461